

Sanchit Sharma

📍 New Delhi, India ✉ sanchitsharma917@gmail.com ☎ +91-7982333535 🌐 GitHub in LinkedIn 📁 Portfolio

Education

Maharaja Surajmal Institute of Technology (GGSIPU), New Delhi Sept 2023 – Present
B.Tech – Electronics & Communication Engineering **CGPA: 9.123 / 10** (Expected June 2027)

Technical Skills

Languages: Java, Python, JavaScript, SQL, HTML/CSS

CS Fundamentals: Data Structures & Algorithms, Object-Oriented Programming (OOPs), Operating Systems, DBMS, Computer Networks, Time/Space Complexity Analysis

Databases: MySQL (relational), MongoDB (NoSQL)

Frameworks & Tools: Node.js, Express.js, React.js, REST APIs, Git/GitHub, Scikit-learn, Pandas, NumPy

Experience

Research Intern — *DRDO, Ministry of Defence, Govt. of India* | New Delhi June 2025 – July 2025

- Designed and implemented **performance-critical algorithms in Java** for defence systems analysis at ISSA, applying Big-O complexity analysis to optimize for correctness and runtime efficiency under strict reliability constraints.
- Translated ambiguous, broadly-defined mathematical and domain-specific requirements into clean, modular, maintainable code — operating in a high-stakes environment with zero tolerance for failure.

Teaching Assistant, DSA & Web Development — *Apna College* | Remote June 2025 – Nov 2025

- Mentored **5,000+ students** on data structures, algorithms, and full-stack web development; resolved 100+ technical doubts weekly, consistently reinforcing problem-solving from first principles and time/space trade-off analysis.

Projects

Finovate — Full-Stack FinTech Application | **React.js, Node.js, Express.js, MongoDB, REST APIs** [Live Demo](#)

- Built a scalable full-stack financial platform with real-time stock charts, secure user authentication, dummy trade execution, and AI-powered market insights serving concurrent users.
- Architected a three-tier system — React front-end, RESTful Node.js/Express.js service layer, and a MongoDB data layer — designed as decoupled, independently scalable components.

DSA Revision Tracker | **JavaScript, Node.js, Express.js, MongoDB** [Live Demo](#)

- Architected a full-stack spaced-repetition system to optimize personal DSA retention, implementing a scheduling algorithm that resurfaces problems at increasing intervals based on recall confidence.
- Built a Kanban-style progress board, gamified streak tracking, and JSON/CSV import-export for portability; deployed a MongoDB/Express REST API backend on Render for persistent multi-device state.

SmartAttend — AI Attendance System | **Python, dlib, SVM, Resemblyzer, Librosa, Supabase** Currently in progress

- Built a dual-mode biometric attendance system: 128-d facial embeddings (dlib) classified via SVM, and speaker verification using Resemblyzer voice embeddings with Librosa-based audio segmentation for bulk voice roll calls.
- Designed a Supabase (PostgreSQL) backend with bcrypt-secured teacher/student roles and QR-code-based self-enrollment; building out the full pipeline as a Streamlit app.

Research & Achievements

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- Published Research Paper** — **NCI-TIDE 2025:** Conducted independent research, authored and presented a peer-reviewed paper at a national-level conference. Dec 2024
 - DSA with Java Certification** — Apna College: Advanced data structures, algorithms, graph theory, dynamic programming, and competitive problem-solving in Java. Oct 2025
 - Full Stack Web Development** — Apna College: End-to-end MERN stack application development and deployment. Dec 2024
 - Microsoft Student Chapter** — Web Developer, MSIT (Sept 2024 – May 2025)
 - Student Representative**, Grievance Redressal Committee, MSIT (Aug 2025 – Present)